A33: The 35-component "law vector" (Residual-Law B3) and the χ^3 sector

For: Arvin Hampton (539 Labs) · 2026-10-22, about 10:36 AM CDT.

Folder: `reports/s_family_paths/A33_law_vector_chi3/`

- Scripts: `law.py`, `module.py`, `to_rho7.py`, `spec_check.py`, `sectors.py`, `exact_z7.py`.
- Outputs: `weil_fit.json`, `module.json`, `module.npz`, `to_rho7.json`, `spec_check.json`, `sectors.json`, `exact_z7.json`, `g_K*.npy`.
- Preregistration: `prereg.md`, SHA-256 `b5bff8ae8652c9e86bf1410e1f8890a3126e1d0d3fd7721138b5c11048e0752c`. It was hashed at 10:28 AM CDT, before any computation, and re-verified when this report was written.

Ground rules

- CORE_FREEZE: H_s was not redefined and certifier code was not touched.
- A32 outputs, RTA_`prereg/results/run_all.py` (EC, the source of ρ_7|W, E and C) and `weil_family` were imported read-only.
- New files only. Nothing was pushed or sent. No TPC or RH claims.
- **Cat A** = exact or proved. **Cat B** = floating point or interpretation.

Verdict: does not land (Cat B for the full statement; the key vanishing is Cat A)

- **Exact (Cat A):** every candidate law vector built from the source's profiles is **even under the local reflection Z_7**.
- So its $\rho_5^- \otimes \rho_7^-$ component, which A32's χ^3 map Φ needs, is identically zero.
- The one exception is the (a,a) characteristic K1. That candidate is identically zero numerically (Cat B) and does not appear in the source's laws.
- **Floating point (Cat B, large gaps):** the profiles do not form a ρ_{35}-module at all.
- Together with their S and T translates, they span a **36-dimensional irreducible** $SL_2(Z)$-module V.
- **V ≅ E**, the 36-dimensional part of ρ_7|W, **with no twist**.
- **Hom(V⊗χ^k, C) = 0 for all 12 k** (and for duals). So neither law·η^6 nor law/η^6, nor any other η^{2k} twist, reaches C.

1. Where the law is defined (sources)

- **Primary source.** `docs/.../Bridge_C_Residual_Law_2026-10-17/2026-10-17-Bridge-C-Residual-Law.md`, line 173, **B3 "Vector-valued escape"**. It says:
- over the 35 characteristic components (Zwegers Cor. 2.9(5) shape), the leading profiles G_μ(x) should transform by the weight-1 Weil representation;
- the expected law covers **the singular parts only**, and the cone sums themselves stay non-modular;
- status: **"Not attempted"**.
- **Other places the same item appears.**
- Line 31 and next step (b) at line 196 of the same file.
- The same text is filed as **B2** in `CatB_Notes_Bridge_C_Residual_Law_2026-10-17.md`.
- `ERRATUM_Bridge_C_tminushalf_Coefficient_2026-10-17.md` E3: the leading order was later checked.
- `P_s_Structure_2026-10-17/2026-10-17-P_s-Structure.md` §A5, `weil_test.py`: the S-law and T-law were checked at 27 + 6 points. The scripts there already speak of a **"36-component family"**.
- RTA_prereg Amendment 2 and A32 only cite it.
- **What the law is.** It is **not a q-series and not an eta or theta quotient.**
- Its components are functions on Q: G_{α,β}(x) = mean over $\nu \in \alpha + Z^2$ of $e(x \cdot Q(\nu) + B(\nu,\beta))$, with $Q(\nu) = \tfrac{1}{2}\nu^T A \nu$ and A = [[2,7],[7,7]].

- Each one is the coefficient of the 1/t singularity of a cone sum at q = e(x)e^{−t}.

- The law expresses weight 1 in "profile form": g(−1/x) = x^{−1}·M_S·g(x) and g(x+1) = M_T·g(x).
- **There is no principal part and no Fourier expansion.** Baker's theorem excludes an exact holomorphic vector-valued law with algebraic data (P_s ledger, A5).
- So item (2) of the brief, "q-expansions", does not apply. What can be built exactly is the finite Gauss-mean value at each rational x.
- **What the source leaves open.** It does not say which 35 pairs (α,β) make up the vector. The preregistered candidates, with μ ∈ D = $A^{-1}Z^2/Z^2$ and the source's characteristic a = (7/10, −1/5), are:
- K1 = G_{a+μ, a}
- K2 = G_{a+μ, 0}
- K3 = G_{μ, a}
- K4 = G_{μ, 0}
- Fact (Cat A): a ≡ ½A^{-1}A* mod $A^{-1}Z^2$.

2. Construction and transformation tests

- **How the values were computed.** Each G is an exact finite mean over a 4m × 4m box. The phases are integers mod 9800m, so no phase rounding occurs; the exponentials themselves are evaluated in floating point.
- Coverage: 216 rationals h/m with m ≤ 15 (138 rationals with m ≤ 12 in `law.py`).
- **Each candidate against $\rho_{35}\otimes\chi^k$** (A32 labels; ε = i; all 12 k; ρ and dual; weight factor x^{±1}), from `weil_fit.json`:
- **K1:** identically 0 (|g| ≤ $4\cdot10^{-16}$). It trivially satisfies every law.
- **K4:** the S-law holds as $\rho_{35}\otimes\chi$, with error $7\cdot10^{-15}$ and weight factor x^{−1}. **The T-law fails** (error 1): T moves the twist β from 0 to the class of a.
- **K2, K3:** they fail both laws.
- **None of the 35-vectors is a $\rho_{35}\otimes\chi^k$ vector.** This is expected for the **odd** lattice A_7. As with θ_2, θ_3, θ_4, T permutes the characteristic classes: K4 →T K3 →S K2.
- **The module V** (`module.py`).
- The 105 functions K2∪K3∪K4 span a space of **dimension 36**, with singular values 1.57 against $3\cdot10^{-14}$.
- The space is closed under S and T: the out-of-span residual is $4\cdot10^{-15}$.
- The fitted M_S and M_T satisfy $S^4 = 1$ and $(ST)^3 = S^2$ to $2\cdot10^{-14}$.
- dim End(V) = 1, so V is irreducible (Cat B).

3. Decomposition and the χ^3 map

- **Exact (Cat A), from `exact_z7.py`, all in rational arithmetic.**
- γ = [[−1,−7],[2,13]] satisfies $\gamma^T A\gamma = A$ and det γ = 1.
- γ^{-1} acts on D **exactly as the local reflection Z_7**: identity on D_5 and −1 on D_7.
- $\gamma^{-1}a - a = (7, -1) \in Z^2$.
- Changing variables ν ↦ γν gives G_{α,β} = G_{γ^{-1}α, γ^{-1}β}. Shifting β by λ ∈ $A^{-1}Z^2$ multiplies G by e(B(α,λ)).
- Checking these phases for all 35 μ gives **g_μ = g_{Z_7μ} exactly for K2, K3 and K4**. So each of these has **zero Z_7-odd part, and in particular zero $\rho_5^-\otimes\rho_7^-$ part**.
- For K1 the phase is ½, so K1 is exactly Z_7-odd. K1 is nevertheless 0 on every test point (Cat B); this was not proved in general.
- **Sector norms (Cat B), from `sectors.json`, maximum over the test rationals.**

candidate	(+,+)	(+,−)	(−,+)	(−,−)	max|Φ g|
K2	1.0	3e−17	0.95	3e−17	2e−16
K3	1.0	6e−17	0.95	2e−17	2e−16
K4	1.0	6e−17	9e−17	3e−17	2e−16
K1	4e−16	2e−17	2e−16	1e−17	1e−17

candidate (+,+) (+,−) (−,+) (−,−) max|Φ g|

- **Module level (Cat B), from `to_rho7.py` and `spec_check.py`.**
- Hom(V⊗χ^k, X) was computed for X ∈ {W, E, C}, their duals, all 12 k, and both transpose conventions.
- The **only** non-zero result is **k = 0 into E (and hence W), with dimension 1**.
- Every χ^k twist into C is 0, **including χ^3** (the η^6 multiplier, weight 3) **and χ^9** (the η^{-6} direction).
- V has 36 distinct T-eigenvalues (in units of 1/140), and they **coincide exactly with those of E**.
- Smallest singular values for Hom(V, E): $3\cdot10^{-15}$, then 0.297.
- Hom(V, ρ_{35}-sector⊗χ^k) = 0 for all 4 sectors × 12 k × {ρ, ρ*}.
- **About weight.** A32's Φ intertwines $\rho_{35}\otimes\chi^3$, and χ^3 is the multiplier of η^6 (weight 3).
- The law has profile weight 1, so a twist law·η^{±6} would have weight 1 ± 3 as a representation-theoretic object.
- **As actual functions this twist is undefined:** η vanishes radially at every rational, so η^6 times a 1/t profile has zero leading profile.
- At the representation level, every twist is covered by Hom(V⊗χ^k, C) = 0 above. Weight compatibility therefore cannot rescue a landing.

4. Verdict and reasons

Does not land.

1. **(Cat A)** Every non-zero candidate 35-vector built from the source's profiles has an exactly vanishing $\rho_5^-\otimes\rho_7^-$ component. The reason is the lattice automorph γ, which acts as Z_7 on D and fixes the characteristic.
2. **(Cat B)** The law is not a $\rho_{35}\otimes\chi^k$ object. Its $SL_2(Z)$-span is a 36-dimensional irreducible V ≅ E, untwisted, and no twist χ^k maps it to C.

What is not claimed

- A general proof of the Weil law for G at all x. The source has it as Cat B; here it is Cat A only per tested point, and the module matrices are fitted.
- An exact (cyclotomic) certificate of V ≅ E.
- K1 ≡ 0 in general.
- Any statement about cone sums beyond leading order.

Possible follow-up: an exact certificate of V ≅ E. Write M_S and M_T in closed form from the S-law and T-law, then rerun A32's mod-p rank method.

Ledger suggestion (Cat B): "The B3 leading-order law vector lives in E (≅, untwisted, 36-dim), not in C; its $\rho_5^-\otimes\rho_7^-$ part vanishes exactly by the automorph γ."